# Agapito Emanuele Santangelo



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## RESEARCH PROFILE

Applied economist working at the intersection of **causal inference**, **digital trace data** and **geospatial / Earth observation data** for the evaluation of public policies, with active research lines on digital transformation, institutional quality, and innovation. Recent work: the MESWI, an equitable and sustainable well-being index for all Italian municipalities (49 indicators, 12 domains).

**Methods:** Causal inference (*generalized and staggered Difference-in-Differences, panel data econometrics*); machine learning and Natural Language Processing (NLP) (*text mining, embeddings, supervised classification*); large-scale digital trace data (*web scraping, social media, high-frequency data*) integrated with administrative data; geospatial and satellite data (*Google Earth Engine, night-time lights, land cover, NDVI, climate data, spatial identification and inference*).

**Geospatial data infrastructure (since 2024):** panel database of all 7,900 Italian municipalities from 20 families of satellite and administrative sources (VIIRS night lights, GHSL, WorldCover, Hansen GFC, MODIS, Sentinel-5P, ERA5-Land, AlphaEarth embeddings) via the Earth Engine API.

**Web GIS and dashboards** (portfolio): the Night Lights Observatory, an interactive map of NASA Black Marble night-time radiance for all 7,903 Italian municipalities, 2012–2024, with growth and decline by inner-area class (MapLibre GL, own data pipeline); the Climate Observatory, 75 years (1951–2025) of ERA5-Land temperature, precipitation and snowfall for every municipality as anomalies against 1961–1990, with warming by elevation band and region; the Reach Observatory, car access to hospitals, upper-secondary schools and railway stations for every municipality, routed with OSRM on OpenStreetMap; a story map on the 2023 Emilia-Romagna flood, from Maxar pre/post imagery and a calibrated sediment mask validated against Copernicus EMS to building exposure and house-price effects by OMI zone; with Instant Analytics, the MESWI territorial observatory (7,899 municipalities) and a daily-updated dashboard on fuel prices across Italian provinces from MIMIT Osservaprezzi data.

## EDUCATION

**PhD in Economics (XXXVIII Cycle), Doctor Europaeus mention** – *University of Molise* — *Nov 2022 – Apr 2026*
Thesis: *Shocks, Public Policies, and Technological Progress: Four Essays on the Drivers of Innovation.*
Visiting PhD Researcher at Universitat Politècnica de València (UPV), Spain, Jan–Jul 2024 (*host: Prof. Josep Domenech*).
*Supervisor: Prof. Giuliano Resce*

**MSc in Management, Entrepreneurship & Innovation, summa cum laude** – *University of Molise* — *Sep 2017 – Apr 2019*
Thesis: *La relazione tra vincoli finanziari e innovazione: un'analisi su un campione di imprese* (The relationship between financial constraints and innovation: an analysis of a sample of firms).
*Supervisor: Prof. Filomena Pietrovito*

## PUBLICATIONS

- Carneiro, B., Resce, G., Caravaggio, N., **Santangelo, A. E.**, Ruscica, G., … & Coffey, K. (2025). *Text mining and machine learning reveal global determinants of food insecurity.* **Scientific Reports**†, 15(1), 36709. [Link]
- Pastore, L., Corvo, L., **Santangelo, A. E.**, Resce, G., & Lelo, A. (2025). *Measuring sustainability impacts through SROI: a case study of the Italian glass industry's contribution to the SDGs.* **Accounting, Auditing & Accountability Journal**†, 38(5), 1483–1498. [Link]
- **Santangelo, A. E.** (2023). *The relationship between financial constraints and innovation, with a supportive analysis on a sample of firms.* **Journal of Academic Research in Economics**, 15(3), 523–544. [Link]
- Pietrovito, F., & **Santangelo, A. E.** (2022). *Divario Nord-Sud: un focus sul Molise.* **GLOCALE**, 18(19). [Link]

## WORKING PAPERS

- Caravaggio, N., Resce, G., & **Santangelo, A. E.** (2025). *EU Cohesion Policy and Digital Public Services* **(Job Market Paper)**. ESDP No. 25100, University of Molise. [Link]
  **Under review**, *Technological Forecasting and Social Change.*
- Caravaggio, N., Resce, G., & **Santangelo, A. E.** (2026). *The Italian Municipality Equitable and Sustainable Well-being Index (MESWI).* arXiv preprint arXiv:2608.26426. [Link]
- Caravaggio, N., Resce, G., & **Santangelo, A. E.** (2026). *Digital Investment and the Evolution of Government Services.* ESDP No. 26104, University of Molise. [Link]
  **Under review**, *Structural Change and Economic Dynamics.*
- Bergantino, A. S., Gabrielli, L., Resce, G., & **Santangelo, A. E.** (2026). *Competition Without Punctuality Gains: Evidence from Italian High-Speed Rail.* SSRN Working Paper. [Link]
  **Under review**, *Transportation Research Part A: Policy and Practice.*
- Pietrovito, F., Rancan, A., Resce, G., & **Santangelo, A. E.** (2026). *Do Fiscal Constraints Affect Health Inequality Research? A Bibliometric Perspective.* ESDP No. 26102, University of Molise. [Link]
- Cicatiello, L., de Angelis, M., Gaeta, G. L., Prota, F., & **Santangelo, A. E.** (2025). *Do Mobility Incentives Improve Research Performance and Career Outcomes? Evidence from Italy's AIM Program.* SSRN Working Paper. [Link]

## REFEREED CONFERENCE PROCEEDINGS

- Cebrián Cerdá, E., **Santangelo, A. E.**, & Domenech, J. (2025). *Are people losing interest in environmental issues? Insights from Google Trends.* CARMA 2025, Rome, 2–4 July 2025. [Link]
- Nigro, I., **Santangelo, A. E.**, & Modina, M. (2025). *Detecting potential fake innovators using web scraping: the case for Italian startup.* CARMA 2025, Rome, 2–4 July 2025. [Link]
- Caravaggio, N., Di Renzo, G., Fanelli, L., Resce, G., & **Santangelo, A. E.** (2024). *In What is Europe Investing? A Text Mining Approach on Cohesion Projects.* CARMA 2024, Valencia, 26–28 June 2024. [Link]
- Carneiro, B., Resce, G., Tucci, G., Ruscica, G., Caravaggio, N., Fanelli, L., **Santangelo, A. E.**, & Cruciata, P. (2024). *A multi-platform framework for nowcasting social phenomena: a case study for food insecurity.* CARMA 2024, Valencia, 26–28 June 2024. [Link]

## RESEARCH & PROFESSIONAL EXPERIENCE

**Post-doctoral Researcher – SAGE-Rise project (WP5, Data-driven Technologies for Evidence-based Policies)** – *University of Molise* — *Sep 2026 – Present*
*Technological solutions for active ageing – validation in real-world settings* (MUR D.D. 307/2025, SAGE-It programme, 24 months): territorial dashboard on access to services in ageing communities, indicators of spatial and social vulnerability, web scraping and NLP, predictive models of needs.
*Supervisor: Prof. Giuliano Resce (WP5 leader)*

**Research Scholarship Holder (*borsa di ricerca*)** – *University of Molise, in agreement with Regione Lazio – Nucleo regionale Conti Pubblici Territoriali* — *May 2025 – Jul 2026*
*Assessing the economic sustainability of Lazio municipalities in the light of the Territorial Public Accounts (CPT)* (FSC 2014–2020, Regione Lazio–UNIMOL protocol, DGR 837/2024): integration of CPT and demographic data, municipal sustainability and territorial cohesion indicators for constitutionally guaranteed services (health, education, mobility), impact of public investment, scenarios on ISTAT demographic projections; partly based at the Nucleo CPT in Rome.
*Scientific lead: Prof. Cecilia Tomassini*

**Research Fellow (*assegno di ricerca*, art. 22 L. 240/2010) – PRIN 2022 Project** – *University of Molise* — *May 2024 – May 2025*
*Reconciling efficiency and equity in the Italian healthcare system after the pandemic crisis* (PRIN 2022, MUR – NextGenerationEU, Dept. of Economics): efficiency, equity and territorial disparities of the Italian healthcare system after the pandemic.
*Principal Investigator: Prof. Giuliano Resce*

**Research Assistant – Food Insecurity Trends Project** – *University of Molise* — *Aug 2023 – May 2024*
Project for the International Center for Tropical Agriculture (CIAT): data-driven analysis of food insecurity drivers and hazards; results published in *Scientific Reports* (2025).
*Supervisor: Prof. Giuliano Resce*

**Junior Expert in Policy Evaluation** – *Eutalia S.p.A.* — *Apr 2023 – Dec 2023*
Evaluation plan and impact evaluation of *PON Research and Innovation 2014–2020.*
*Coordinator: Prof. Francesco Prota*

## TEACHING

**Teaching Assistant** – *University of Molise* — *2022 – Present*
Since 2022: Monetary and Financial Economics and Policy; Macroeconomics; Industrial Economics and Game Theory; Microeconomics; Cost-Benefit Analysis; Economic Policy.

Since 2025: Applied Health Economics; Public Economics.

## CONFERENCE PRESENTATIONS & WORKSHOPS

- **SIE 2026** – 67th Annual Scientific Meeting of the *Italian Economic Association (Società Italiana di Economia)*, Università di Cagliari, 22–24 October 2026. *Presenter.*
- **CARMA 2026** – 8th Int. Conf. on Advanced Research Methods and Analytics, Universitat Politècnica de València, 1–3 July 2026. *Presenter.*
- **AISRe 2026** – XLVII Annual Scientific Conference of the *Italian Regional Science Association* (RSAI Italian Section), Università di Firenze, 24–26 June 2026. *Presenter.*
- **Understanding and Addressing Digital Inequalities** – Annual Scientific Conference on Digital Democracy, *Centre for a Digital Society, European University Institute (EUI)*, Florence, 20–21 November 2025. *Presenter.*
- **SIE 2025** – 66th Annual Scientific Meeting of the *Italian Economic Association (Società Italiana di Economia)*, Università di Napoli Parthenope, 23–25 October 2025. *Presenter.*
- **AISRe 2025** – XLVI Annual Scientific Conference of the *Italian Regional Science Association* (RSAI Italian Section), Università "G. d'Annunzio" Chieti-Pescara, 10–12 September 2025. *Presenter.*
- **CARMA 2025** – 7th Int. Conf. on Advanced Research Methods and Analytics, Rome, 2–4 July 2025. *Presenter.*
- **CARMA 2024** – 6th Int. Conf. on Advanced Research Methods and Analytics, Valencia, 26–28 June 2024. *Presenter.*

## ACADEMIC SERVICE

**Referee for:** The Annals of Regional Science† · Italian Economic Journal† · Research in International Business and Finance† · PLOS ONE† · Journal of Infrastructure, Policy and Development · The B.E. Journal of Economic Analysis & Policy†.
† ANVUR Class A journal (Area 13).
**Conference organization:** CARMA 2025 (Rome) · CARMA 2024 (Valencia) · HEAd'24 – 10th Int. Conf. on Higher Education Advances.

## POLICY REPORTS & DISSEMINATION

- Carneiro, B., Resce, G., Fannelli, L., Ruscica, G., Gomes, F., & **Santangelo, A. E.** (2024). *Climate security narratives among ClimBeR partners: A web analytics approach to measure influence and reach of ClimBeR's WP2.* Report, CGIAR Climate Resilience Initiative (ClimBeR). [Link]
- *PON Research and Innovation 2014–2020 – Final Evaluation Report.* [Link]
- Eticaeconomia: *In cosa investe l'Unione Europea? Un'analisi di Text Mining sui progetti di coesione.* [Link]
- Eticaeconomia: *Il PNRR nella comunicazione istituzionale dei comuni.* [Link]

## ADVANCED TRAINING

**Geospatial & Earth Observation Data**

- **EO Summer School** – *Satellite Data in Agricultural and Environmental Economics* (University of Milan, 2026): remote sensing and Google Earth Engine (Land Economics Group, University of Bonn).
- **NASA ARSET** – *Fundamentals of Remote Sensing* and *Hyperspectral Data for Land and Coastal Systems* (NASA Applied Remote Sensing Training Program, 2026).
- DataCamp – *Working with Geospatial Data in Python* (GeoPandas, spatial joins, projections, overlays, raster data; 2026).

**Econometrics, Causal Inference & Policy Evaluation**

- 6th Summer School in Advanced Economics – *Modern Causal Inference* (Prof. J. M. Wooldridge, Prof. P. Sant'Anna; University of Ioannina, Spetses, 2025).
- LSE Summer School – *Advanced Econometrics* (Dr. M. Schafgans, Dr. M. Gmeiner, 2025).
- SIdE – *Introductory Econometrics* (Prof. G. Calzolari, 2024) and *Panel Data Econometrics* (Prof. M. E. Bontempi, Prof. R. Golinelli, Dr. I. Mammi, 2024).
- *Evaluation of Development Policies* (Prof. P. Montalbano, Prof. F. C. Rosati, Prof. C. Pietrobelli, 2023).

**Data Science, Machine Learning & Big Data**

- SoBigData Academy Camp – *Data Science, Machine Learning, Data Ethics and Social Data Analysis* (SoBigData Research Infrastructure, 2026).
- LSE Summer School – *Applied Econometrics and Big Data* (Prof. T. Komarova, Dr. M. Schafgans, 2025).
- DataCamp – *Web Scraping in Python* and *Web Scraping in R* (2026).
- SAS Training – *Data organization and loading for SAS Visual Analytics* (2025).
- Harvard Online *Professional Certificate in Data Science* (2023).

## SKILLS, LANGUAGES & AWARDS

**Programming & tools:** R, Stata, Python, Google Earth Engine, ArcGIS, QGIS, Web GIS, LaTeX.
**Languages:** Italian (native), English (C1), Spanish (B1).
**Awards:** *America Giovani* Prize, Fondazione Italia USA – top 1,000 Italian graduates for academic excellence (2019).
**Memberships:** Società Italiana di Economia (SIE) · Associazione Italiana di Scienze Regionali (AISRe).